

PsyBU Math Notation

When completing learning assessments on Canvas, there are many questions where you will be asked to provide an answer by typing a formula or an equation. Answers to these questions must be typed using a specific type of notation that I call “PsyBU math notation.” This is a type notation that is unique to this class (“PsyBU” stands for Psychology at Baylor University and it is also the name of a website where statistics music videos are posted), and it is a notation where everything can be typed using the standard characters on a basic keyboard. By using this notation, it is possible for answers to be graded automatically on Canvas.

In general, PsyBU math notation involves: using a caret symbol (^) to indicate superscripts, using an underscore (_) to indicate a subscripts, keeping in mind the standard order of operations, and writing formulas or equations with no spaces using standard un-formatted characters from a basic keyboard.

The specific procedures for using PsyBU math notation are as follows:

1. Use the following terms and symbols to write equations:

_	Underscore indicates a subscript. For example, x_i should be written x_i.
^	Caret symbol indicates a superscript. For example, x^2 should be written x^2.
sqrt(x)	Typing “sqrt” followed by something enclosed in parentheses indicates taking the square root of whatever is enclosed in parentheses.
sum(x)	Typing “sum” followed by something enclosed in parentheses indicates taking the sum of whatever is enclosed in parentheses. It is a replacement for the Greek “Σ” symbol (which is the upper-case Greek letter sigma typically used as a summation sign).
(x)(y)	Two adjacent terms each enclosed in parentheses indicates that the terms should be multiplied.
/	A forward slash indicates division.
+	A plus symbol indicates addition.
-	A single dash symbol indicates subtraction.
bar	Type the word “bar” <i>after</i> a letter (with no spaces) to indicate that the letter should have a bar over it. For example, $\bar{x}$ should be written xbar.
hat	Type the word “hat” <i>after</i> a letter (with no spaces) to indicate that the letter should have a hat over it. For example, $\hat{x}$ should be written xhat.
Greek letters	
mu	Type “mu” to indicate μ (the Greek letter mu).
beta	Type “beta” to indicate β (the Greek letter beta).
rho	Type “rho” to indicate ρ (the Greek letter rho).
sigma	Only type “sigma” to indicate σ , which is the lower-case Greek letter sigma that is NOT a summation sign. (Use “sum” for the upper-case Greek Σ letter.)
alpha	Type “alpha” to indicate α (Greek letter alpha).

2. **NEVER, EVER** type any **spaces**. Each formula or equation must be a string of characters with NO spaces. If you insert a space anywhere in a formula or equation, Canvas will score it as being wrong (although it *might* ignore a space typed at the very beginning or very end of an answer).
3. Use only un-formatted characters from a basic keyboard. Do NOT use formatting to create characters that appear as superscripts or subscripts. Do NOT use any kind of equation editor. Do NOT use formatting to change the font. For example, do NOT use a font that prints Greek characters. Do NOT change the font to bold or italic.

4. When writing equations, keep in mind the standard order of operations:

Parentheses (simplify inside the parentheses first)

Note: this includes performing the following operations: $\text{sum}(x-\bar{x})$ and $\text{sqrt}(x)$.

Exponents

Multiplication and Division from left to right

Addition and Subtraction from left to right

Regarding order of operations, note that, if a fraction has components in the numerator or in the denominator that are summed or subtracted, these components need to be placed in parentheses.

For example, the formula $\frac{a+b}{n-1}$ would be written $(a+b)/(n-1)$.

Important Tips

5. It is highly recommended that you first write out the equation or formula on a piece of paper and then translate the equation into PsyBU math notation. It could be tricky to hold the equation in your mind as you translate it into PsyBU notation, and you will decrease the risk of translation errors if you first write the equation on a piece of paper.
6. **All** the questions that require use of PsyBU notation are contained within the *Lecture Study Guide*, and the *Lecture Study Guide addendum* provides a list of all the correctly spelled formulas and equations that are needed to answer these questions. So, if you see a question on the *Lecture Study Guide* that asks for a formula or equation, make sure your answer to the question matches one of the options listed in the addendum. This will ensure that you have the correct PsyBU notation for all the questions you may encounter on any learning assessment.
7. To ensure that you receive credit, it is essential to be careful and precise when using PsyBU notation. Notably, procedures will be used to reduce the likelihood of students being penalized for minor typos when using PsyBU notation. Whenever possible, Canvas will be programmed to accept as correct alternate versions of formulas or equations that contain common typos but still reflect a full and accurate understanding of the formula or equation at hand.

Other Notes

8. When writing formulas and equations in PsyBU math notation, write all letters as lower-case letters. This is not a crucial rule because Canvas ignores whether letters are upper- or lower-case when grading. However, in the answer key, all the correct equations are written using only lower-case letters, and you will make things easier on yourself if you simply write all letters as lower-case.
9. As a general rule, if you are asked to write an **equation**, it will need to include an **equal sign**. In contrast, if you are asked to write the **formula** for something, you **only** need to write the formula and **not** a full equation with an equal sign.
- For example, if you are asked to write a linear **equation** in which the mean is used to predict scores on x , you would write " $x_i = \bar{x} + e_i$ " and not " $\bar{x} + e_i$." The key thing to note in this example is that the correct version includes an equal sign.
 - In contrast, if asked to write the **formula** for the d , you would write, " $(\bar{x}_1 - \bar{x}_2) / \text{sd_pooled}$ " and NOT " $d = (\bar{x}_1 - \bar{x}_2) / \text{sd_pooled}$." The key thing to note is that because the question is asking for **formula** for d , an equal sign is not necessary. In other words, you do NOT need to type " $d =$ " because it is obvious that your answer is the formula for d .
10. If a character has both a superscript and a subscript, it does not matter which you type first and which you type second. Both versions will be entered as correct answers on Canvas.